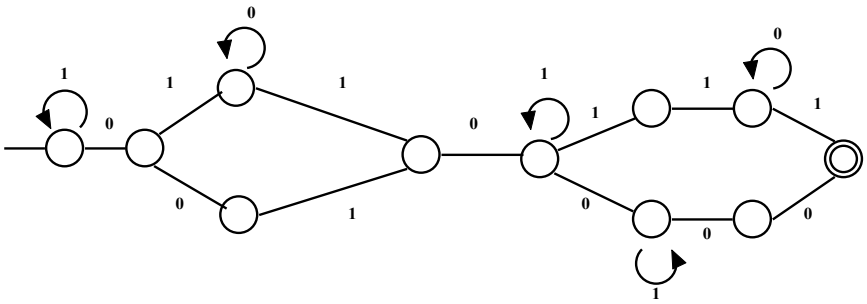
Compiled Problems – FSAs

1. 2002-03

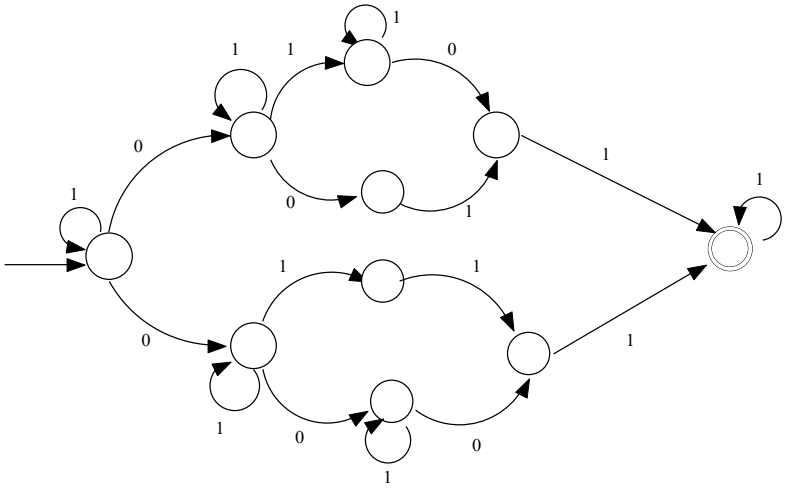


List all of the following regular expressions that represent strings which can be generated by the FSA above.

a. 101101110
b. 1*0(10*1U01) 01*(110*1U01*00)
c. 101010101010
d. 00100100
e 1*010*101*110*1

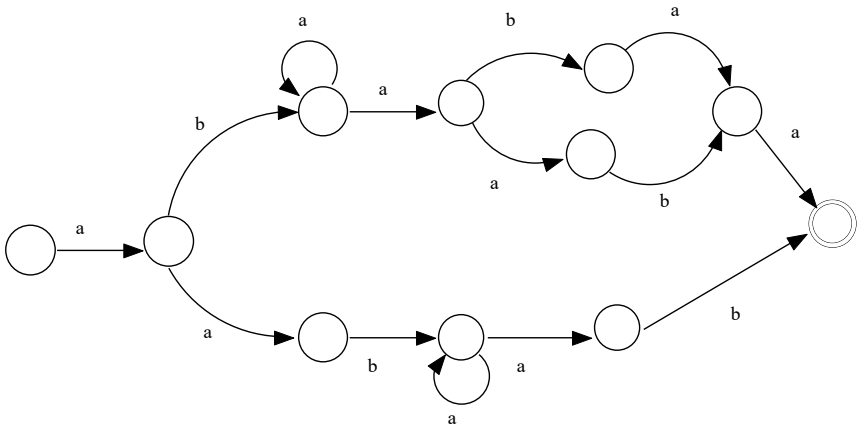
2. 2003-04

What is the difference between the longest and shortest strings that contain exactly 6 1's.



3. 2004-05

What Regular Expression represents this FSA?

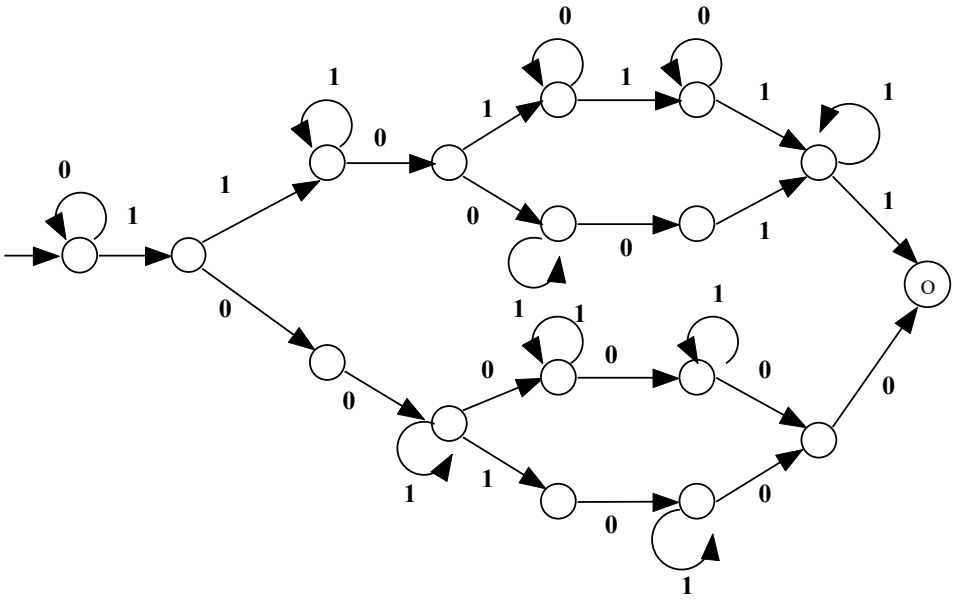


4. 2005-06

Which regular expressions are accepted by the following FSA?

A) 100110110 B) 01101010101

C) 0011100111001 D) 001001110110111100



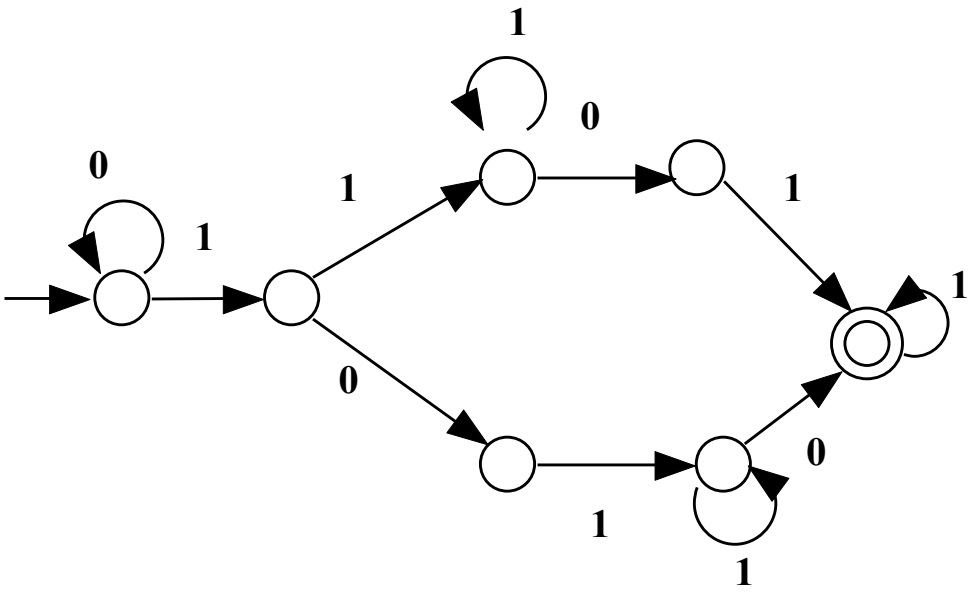
5. 2006-07

Which regular expressions are accepted by the following FSA?

I) 101101101 II) 111100111

III) 1011101111 IV) 1010101

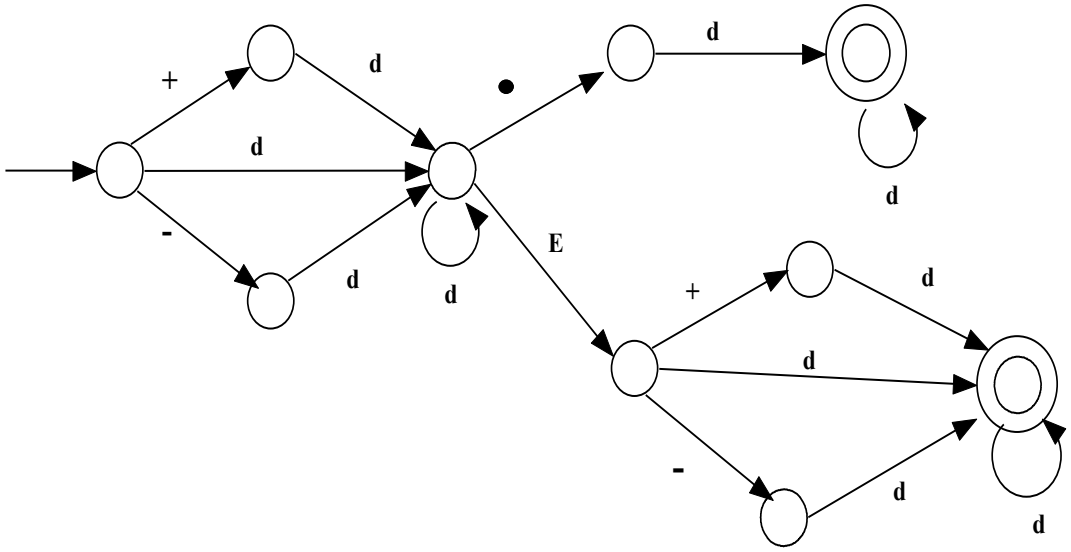
V) 11111111111



6. 2008-09

If d = {0, 1, 2, …, 9}, which of the following strings are represented by the following FSA?

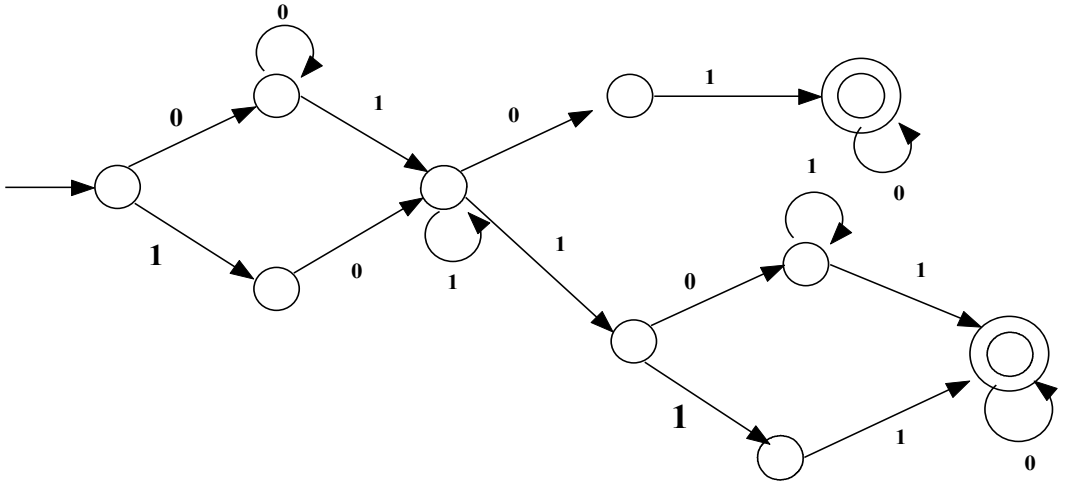
a) 57.4 b) -26E3 c) 42.6E-2 d) -.478 e) +8.35 f) 5.4E-8.3



7. 2009-10

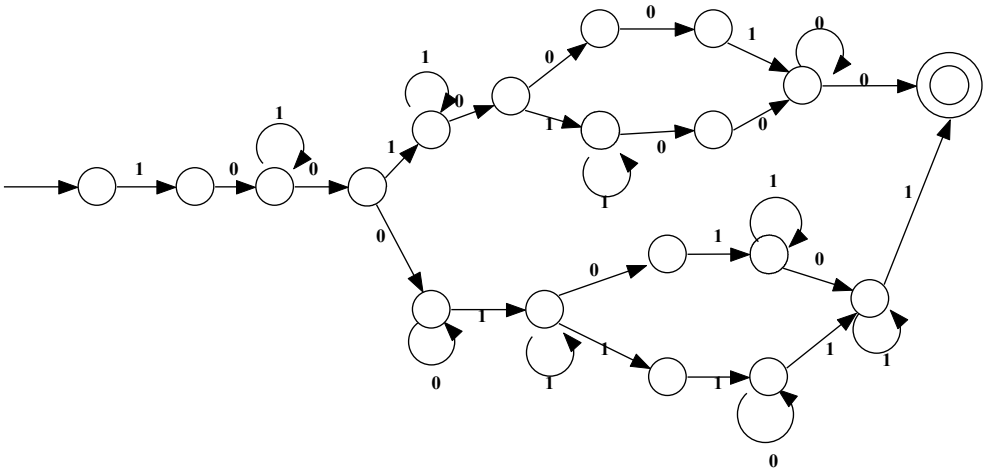
Which of the following strings are represented by the following FSA?

a) 001111 b) 1010101 c) 1011101
d) 100000 e) 1010111 f) 00111011100



8. 2012-13

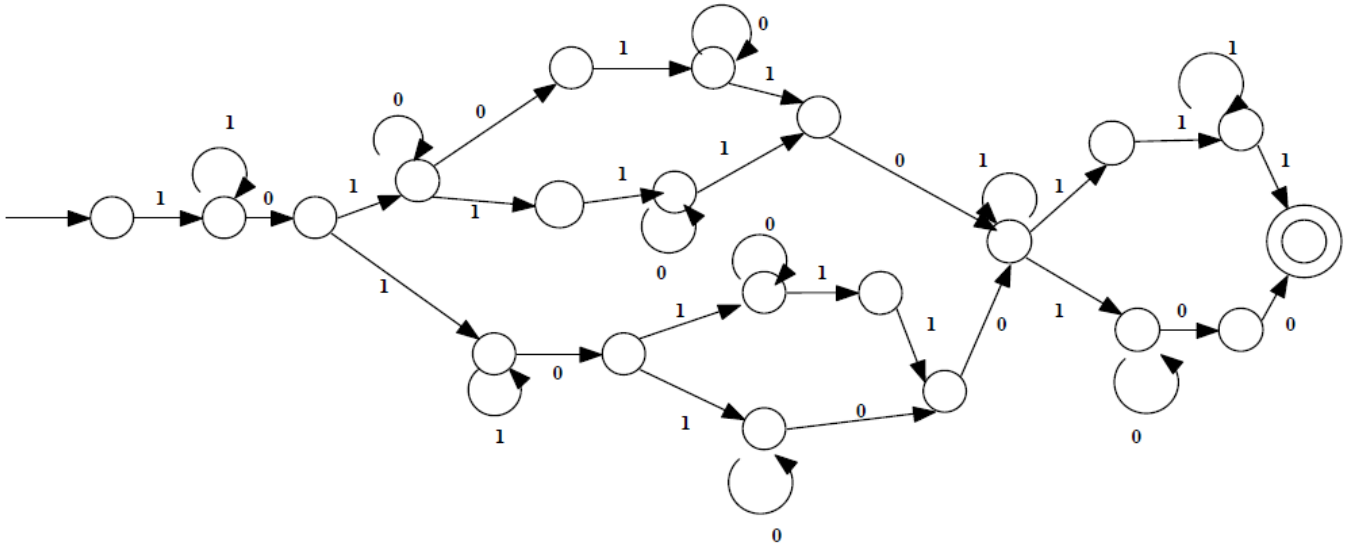
Which of the strings below are represented by the following FSA?



a. 101011001011
b. 101100110101
c. 1010001111001111111
d. 1011100010110010
e. 10000011111111
f. 100101000

9. **2013-14**

Which of the strings below are represented by the following FSA



a. 1010100111
b. 11010110100101
c. 1010011001011111111
d. 1101101011011100
e. 1010100011010
f. 11010001000101111000000000

10. 2014-15

Translate the following FSA into a regular expression.

